3(a)	force × displacement in the direction of the force	**B1**
3(b)(i)	displacement = 4.4×30	**C1**
	work done = $140 \cos 30° \times 4.4 \times 30$	**C1**
	$= 1.6 \times 10^4$ J	**A1**
3(b)(ii)	$p = F / A$	**C1**
	$F = 860 - 140 \sin 30°$ (= 790)	**C1**
	$A = 790 / 2400$ $= 0.33$ m^2	**A1**
3(b)(iii)	$\sigma = F / A$ or $F / \pi r^2$ or $4F / \pi d^2$	**C1**
	$9.6 \times 10^6 = 4 \times 140 / \pi d^2$ $d = 4.3 \times 10^{-3}$ m	**A1**

Question	Answer	Marks
3(c)	$E = ½Fx$ or $½kx^2$ or area under graph	C1
	$(\Delta)E = ½ \times (140 + 210) \times 0.20 \times 10^{-3}$ or $(\Delta)E = (½ \times 210 \times 0.60 \times 10^{-3}) - (½ \times 140 \times 0.40 \times 10^{-3})$ or $(\Delta)E = (140 \times 0.20 \times 10^{-3}) + (½ \times 0.20 \times 10^{-3} \times 70)$ or $(\Delta)E = [½ \times 3.5 \times 10^5 \times (0.60 \times 10^{-3})^2] - [½ \times 3.5 \times 10^5 \times (0.40 \times 10^{-3})^2]$	C1
	$\Delta E = 0.035$ J	A1

Question	Answer	Marks